

Product Requirements Document

Wellfarm — Early Crop Disease and Pest Detection and Management System

SIH Problem Statement: SIH26131

Organization: Government of Maharashtra — Maharashtra State Innovation Society

Category: Software

Theme: Agriculture, FoodTech & Rural Development

Prototype Stage: Smart India Hackathon Round 1

Document Version: Demo Prototype PRD

1. Product Summary

Wellfarm is an India-wide crop disease and pest intelligence platform for farmers, agriculture officials, laboratories and field officers.

The platform begins with an individual farmer scanning an affected crop. It then connects that scan to weather conditions, nearby reports, district-level outbreak patterns, government monitoring and laboratory verification.

Unlike a basic crop-image classifier, Wellfarm is designed as a closed intelligence loop:

1. A farmer submits a crop scan.
2. Wellfarm provides an early probable diagnosis.
3. The farmer receives localized, understandable guidance.
4. The anonymized report contributes to district-level outbreak monitoring.
5. Officials receive regional warnings and recommended actions.
6. Moderate- and high-risk cases can be routed to a compatible laboratory or verification center.
7. Laboratory or field findings return to the system.
8. Verified findings improve future detection and advisory quality.

The Round 1 prototype must make this complete vision visible while clearly distinguishing live, prototype and simulated functionality.

2. Problem Statement

Farmers frequently discover crop diseases and pest infestations after visible damage has already spread.

Agricultural-extension coverage is limited, diagnosis may be delayed, and individual farmer reports are rarely connected to broader regional patterns.

This contributes to:

- Delayed treatment
- Incorrect treatment
- Excessive or inappropriate pesticide use
- Increased cultivation costs
- Pesticide-residue risk
- Preventable yield loss
- Limited access to experts and laboratories
- Poor government visibility into emerging outbreaks
- Weak coordination between farmers, officials and laboratories

Existing crop-diagnosis applications generally focus on helping one farmer identify one image. They do not necessarily connect the diagnosis to nearby reports, live regional conditions, government response and field verification.

3. Product Vision

Wellfarm should become a shared crop-health intelligence platform that:

- Helps farmers identify problems earlier
- Provides safe, localized guidance
- Detects geographically concentrated crop threats
- Relates outbreak patterns to weather conditions
- Helps officials prioritize investigations
- Routes relevant cases to appropriate verification centers
- Learns from laboratory-confirmed and field-confirmed outcomes
- Reduces repeated computation for recurring validated patterns

Wellfarm is a decision-support system. It is not a replacement for agricultural experts, laboratory testing or official pesticide guidance.

4. Target Users

4.1 Farmers

Farmers need to:

- Upload or capture crop photographs
- Provide crop and symptom information
- Share GPS location with consent
- Receive an early probable diagnosis
- Understand confidence and uncertainty
- Receive guidance in a selected Indian language
- See weather-related warnings
- Receive laboratory or expert-referral information
- Track previous scans and referrals
- Contribute anonymized information to community monitoring

4.2 Agriculture Officials

Officials need to:

- Monitor crop-health conditions across India
- Drill down from country to state to district
- View outbreak severity
- Filter reports by crop, condition, date and status
- Inspect regional trends
- Review weather–outbreak associations
- Receive system-generated recommended actions
- Monitor laboratory referrals
- Schedule field investigations
- Review monthly reports
- Understand data provenance and model limitations

4.3 Laboratory and Field Officers

Laboratory and field users need to:

- Receive assigned cases
- View case evidence

- See approximate location and weather context
- Record field or laboratory findings
- Confirm or reject a model prediction
- Mark results inconclusive
- Recommend follow-up action
- Submit verified findings into the collective dataset

4.4 Judges and Demonstration Viewers

The prototype needs a guided demonstration mode that explains:

- What is live
 - What is powered by a prototype model
 - What uses simulated data
 - What represents future production architecture
-

5. Supported Platforms

Wellfarm will contain:

1. A mobile-first farmer web application
2. A responsive public Wellfarm website
3. An officials' web dashboard inside the main Wellfarm website
4. A laboratory/field-officer interface inside the same application
5. A guided judge-demo mode

The farmer experience may later become a native mobile application, but the first prototype may be delivered as a responsive web application or Progressive Web App.

Offline operation is not required for the first demo.

6. Initial Crop Coverage

The prototype architecture should support at least these ten crops:

1. Rice
2. Wheat
3. Maize

4. Cotton
5. Sugarcane
6. Soybean
7. Groundnut
8. Tomato
9. Potato
10. Onion

The interface may expose all ten crops while the initial computer-vision prototype supports a smaller validated set of crop-condition combinations.

Unsupported crops or conditions must return an honest unsupported result instead of forcing a diagnosis.

7. Initial Condition Coverage

The proposed first seven crop-condition classes are:

1. Rice — bacterial leaf blight
2. Rice — brown spot
3. Maize — northern/Turcicum leaf blight
4. Maize — fall armyworm damage
5. Tomato — early blight
6. Potato — late blight
7. Wheat — leaf rust

The final seven may change after dataset evaluation.

Additional required classification outcomes:

- Healthy
- Unknown
- Unsupported crop or condition
- Poor-quality image
- Insufficient visual evidence

8. Language Support

The prototype should provide a language-selection system for:

- English
- Hindi
- Marathi
- Bengali
- Telugu
- Tamil
- Gujarati
- Kannada
- Malayalam
- Punjabi
- Odia
- Assamese

The language setting should affect:

- Navigation
- Form labels
- Scan instructions
- Diagnosis explanations
- Safety notices
- Farmer advisories
- Referral information
- Error messages

Text-to-speech is not required.

Audio descriptions are not required.

Interface strings should be stored in maintainable locale files rather than embedded directly into components.

9. Farmer Application Requirements

9.1 Farmer Home

The farmer home screen should contain:

- Current selected language

- Approximate detected area
- Current weather summary
- Primary “Scan crop” action
- Recent scans
- Referral status
- Nearby community warning
- Explanation that anonymized scans help the surrounding farming community

9.2 Crop Image Submission

Farmers should be able to:

- Capture a photograph
- Upload an existing photograph
- Preview the photograph
- Replace the photograph
- Remove the photograph
- See guidance for taking a useful image

The application should validate:

- File type
- File size
- Image availability
- Basic image quality
- Whether the image appears supported

9.3 GPS Location

The application should request GPS permission.

Location should be obtained directly through the device or browser rather than manually entered by the farmer.

The application should handle:

- Permission granted
- Permission denied
- Location timeout
- Low accuracy
- GPS unavailable
- Demo-location fallback

Exact farm coordinates should not be shown on the officials' dashboard.

Analytics should use an approximate geographic cluster, district or similarly privacy-safe area.

9.4 Farmer Input Form

The farmer should provide:

- Crop
- Visible symptoms
- Affected plant part
- Growth stage or approximate plant age
- Approximate affected area
- Whether nearby plants show similar symptoms
- Optional written notes

The farmer should not be required to provide:

- Audio description
- Date symptoms began
- Manual address when GPS succeeds

9.5 Analysis Process

The visible analysis flow should include:

1. Image-quality check
2. Crop compatibility check
3. Computer-vision analysis
4. Weather retrieval
5. Nearby-pattern lookup
6. Solutions-cache lookup
7. Advisory preparation
8. Risk assessment
9. Referral decision

9.6 Diagnosis Result

The diagnosis result should display:

- Probable condition
- Confidence

- Risk level
- Image-quality result
- Model version
- Prototype-model label
- Short explanation
- Immediate safe actions
- What to avoid
- When expert assistance is required
- Weather-related warning
- Verification status
- Data-source/provenance labels
- Community contribution confirmation
- Referral status

The result must explain that it is an early indication rather than a confirmed diagnosis.

9.7 Scan History

Farmers should be able to see:

- Previous scans
- Crop
- Probable condition
- Risk
- Date
- Verification status
- Referral status

10. Computer-Vision Detection Layer

The computer-vision component should:

- Accept a crop image
- Detect whether the image quality is sufficient
- Return ranked candidate labels
- Return calibrated confidence
- Return crop compatibility
- Return an unknown or unsupported outcome

- Return the model version
- Avoid generating treatment advice directly

The LLM must not interpret unstructured model scores without a controlled structured layer.

Round 1 treatment

The prototype may use:

- A small pretrained or fine-tuned model
 - A limited set of supported classes
 - Deterministic mock inference until the model is ready
 - Clearly labeled sample results
-

11. Weather Integration

Weather must be retrieved using the farmer's GPS coordinates.

The prototype should support:

- Current temperature
- Humidity
- Recent rainfall
- Forecast rainfall
- Daily minimum and maximum temperature
- Weather code
- Observation timestamp
- Data-source label

Open-Meteo is the preferred first integration.

The prototype should include deterministic fallback weather if the live API fails.

Historical weather is also required for the simulated weather–outbreak correlation demonstration.

12. Collective Intelligence Layer

Anonymized farmer scans should be grouped into geographic clusters.

The system should calculate or display:

- Number of reports
- Number of unique farms
- Crop
- Probable condition
- Time window
- Geographic concentration
- Verification count
- Recent trend
- Weather suitability

The system should not expose individual farm coordinates to officials.

Round 1 treatment

Use deterministic simulated outbreak records while keeping the visible aggregation and drill-down interactions functional.

13. Weather–Outbreak Correlation

The intelligence layer should compare:

- Temperature
- Humidity
- Rainfall
- Recent weather windows
- Crop reports
- Condition reports
- Geographic clusters
- Verified findings

The goal is to identify weather signatures associated with the appearance or growth of an outbreak.

The output should include:

- Weather variables involved
- Time window
- Report count

- Geographic area
- Strength of association
- Confidence or uncertainty
- Plain-language explanation

The system must not claim that weather caused an outbreak merely because the values are correlated.

Round 1 treatment

Show a defined and explainable simulated example. Do not claim that a production unsupervised-learning system is already operating.

14. LLM Advisory Layer

The LLM advisory system should receive structured information from:

- Computer-vision output
- Weather service
- Regional outbreak service
- Risk calculation
- Approved agricultural guidance
- Solutions cache

It should generate:

Farmer-facing advisory

- Simple language
- Selected regional language
- Immediate safe steps
- Observation instructions
- Escalation conditions
- Uncertainty statement

Official-facing recommendation

- Regional summary
- Evidence used
- Recommended investigation

- Suggested field-survey priority
- Referral information
- Confidence and limitations

The LLM must not invent:

- Pesticide dosages
- Waiting periods
- Regulatory instructions
- Unsupported products
- Laboratory confirmation
- Government endorsement

A deterministic advisory must be available if the live LLM integration fails.

15. Solutions Cache

The system should store sufficiently validated combinations of:

- Crop
- Condition
- Weather pattern
- Geographic context
- Risk level
- Approved advisory
- Verification count

When a similar new case arrives:

1. The system checks the solutions cache.
2. If a validated pattern matches, the previous solution can be reused.
3. If no match exists, the full analysis/advisory pipeline runs.

Visible prototype demonstration

The prototype should show:

- First case using full inference
- Pattern validation
- Solution stored

- Similar second case
- Cache match
- Reduced simulated processing time
- Previous validation count

All prototype timing values must be labeled as simulated.

16. Officials' Dashboard

16.1 National Overview

The dashboard should contain:

- India-wide map
- Every state and union territory
- State severity
- Search
- Crop filter
- Condition filter
- Severity filter
- Date filter
- Data-source filter
- Last-updated timestamp
- Map legend
- Non-map summary

16.2 State Drill-Down

Selecting a state should show:

- District-level map
- Highest-risk districts
- New reports
- Verified cases
- Pending referrals
- Recent changes
- Leading crop-condition combinations

16.3 District Detail

Selecting a district should show:

- Current severity
- Report count
- Unique reporting farms
- Crop breakdown
- Condition breakdown
- Seven-day trend
- Thirty-day trend
- Weather signature
- Confidence distribution
- Verified and unverified reports
- Recommended actions
- Cases
- Referrals
- Data provenance

16.4 Severity Levels

Use:

- Green — low
- Amber — moderate
- Muted red — high

Color must be accompanied by text and icons.

16.5 Severity Calculation

The future severity formula should consider:

- Number of unique farms
- Report volume relative to local crop prevalence
- Recent report growth
- Geographic clustering
- Model confidence
- Independent image count
- Laboratory confirmations
- Weather suitability
- Duplicate filtering

- Report recency

Severity must not be based only on model confidence.

The prototype should provide a “Why this severity?” explanation.

17. Monthly Reporting

The system should generate a monthly report containing:

- Executive summary
- State and district risk rankings
- Leading affected crops
- Leading conditions
- Report trends
- Verified findings
- Weather associations
- Referral activity
- Recommended surveys
- Recommended official actions
- Data limitations

Round 1 treatment

Use a professional report preview generated from deterministic demo data.

Provide a print or download interaction.

Label the report as based on prototype/demo data.

18. Laboratory and Verification-Centre Directory

Laboratory and verification-centre data should eventually be collected from public sources such as:

- ICAR institute directories
- ICAR accredited laboratories
- State Agricultural Universities
- Krishi Vigyan Kendras

- NABL laboratory search
- Relevant government laboratory directories

Data should be collected using a controlled scraper where permitted.

Each record should contain:

- Name
- Organization
- Organization type
- Address
- State
- District
- PIN code
- Coordinates
- Phone
- Email
- Website
- Supported crops
- Supported services/tests
- Accreditation
- Verification date
- Source URL

Not every KVK, institute or soil-testing facility should be treated as a crop-disease diagnostic laboratory.

Each center must have explicit service capabilities.

19. Laboratory Routing

Routing should first determine whether a center can handle the suspected issue.

Routing factors should include:

1. Required diagnostic capability
2. Active/verified status
3. Crop specialization
4. Condition or test specialization

5. State/district availability
6. Geographic distance
7. Workload or availability, when known
8. Case urgency

The closest facility should not be selected if it lacks the correct capability.

Suggested behavior:

- Low risk: monitor, no automatic referral
- Moderate risk: suggest compatible verification center
- High risk: prioritize official and laboratory review
- Low-confidence result: expert review
- Unsupported result: request another scan or manual review

Round 1 treatment

Use a small scraped or curated directory and perform a real distance-and-capability routing calculation where possible.

20. Laboratory and Field Verification

Laboratory or field officers should be able to:

- View assignments
- Open a case
- Inspect the submitted image
- View model findings
- View approximate location
- View weather context
- Record sample collection
- Confirm the predicted condition
- Reject the predicted condition
- Mark the result inconclusive
- Add findings
- Recommend follow-up
- Submit verification

After submission, the prototype should visibly demonstrate:

- Verification status update
 - District severity recalculation
 - Farmer case update
 - Record entering the verified dataset
 - Eligibility for future model training
-

21. Incremental Retraining and Fingerprinting

The production design should periodically prepare verified records for controlled model retraining.

Each record should be fingerprinted.

The system should:

- Process new or changed records
- Skip unchanged records
- Create versioned training snapshots
- Evaluate candidate models
- Compare the candidate against the current model
- Require approval before deployment
- Preserve rollback capability

Round 1 treatment

Show a simulated retraining operations panel containing:

- Last retraining time
- New verified records
- Changed records processed
- Unchanged records skipped
- Candidate score
- Existing model score
- Deployment status
- Rollback status
- Next scheduled retraining

Clearly label this as simulated.

22. Guided Judge Demonstration

The application should include a guided demonstration mode lasting approximately three to five minutes.

Primary scenario

1. A rice farmer near Cuttack uploads a leaf image.
2. GPS identifies the approximate district.
3. Current weather is retrieved.
4. The prototype identifies possible bacterial leaf blight.
5. The farmer receives localized advice.
6. The scan joins an existing district cluster.
7. District severity changes from low to moderate.
8. The officials' dashboard highlights the district.
9. A recent humidity/rainfall association is displayed.
10. A compatible nearby verification center is recommended.
11. A laboratory or field officer opens the case.
12. The finding is confirmed.
13. The district dashboard updates.
14. The verified record becomes eligible for retraining.
15. A second similar case demonstrates a solutions-cache hit.

The guided mode should:

- Explain each step
 - Highlight the relevant interface
 - Provide next/back controls
 - Allow exit at any time
 - Explain which components are live, prototype or simulated
-

23. Data Provenance

The interface should use consistent provenance labels:

- Live weather
- Prototype model
- AI-generated advisory
- Demo outbreak data

- Simulated retraining
- Sample referral
- Verified field result
- Cached validated solution

A transparency page should explain every label.

24. Privacy and Safety

Wellfarm should follow these principles:

- Ask permission before accessing GPS.
 - Avoid exposing exact farm coordinates.
 - Aggregate official-facing information geographically.
 - Separate identifying data from analytical data.
 - Display model uncertainty.
 - Support unknown and unsupported outcomes.
 - Avoid forced diagnosis.
 - Avoid unsourced pesticide advice.
 - Avoid presenting association as causation.
 - Preserve data provenance.
 - Keep credentials outside frontend code.
 - Do not use private farmer images as public demo content.
 - Provide human/laboratory escalation for significant cases.
-

25. Accessibility

The prototype should:

- Meet WCAG AA contrast
- Use keyboard-accessible controls
- Show visible focus states
- Use comfortable touch targets
- Pair colors with text and icons
- Include a map legend
- Include a non-map regional summary

- Support Indian scripts correctly
 - Provide useful loading and error states
 - Respect reduced-motion preferences
 - Remain usable on mobile, tablet and desktop
-

26. Visual Design Direction

Wellfarm should appear:

- Trustworthy
- Calm
- Agricultural
- Practical
- Modern
- Human
- Suitable for farmers and public officials

Preferred visual characteristics:

- Warm off-white backgrounds
- Deep leaf green
- Muted moss and sage
- Soil-brown accents
- Amber warnings
- Muted brick-red high-risk states
- Clear typography
- Restrained borders
- Subtle shadows
- Strong information hierarchy
- Functional data visualization

Avoid:

- Neon colors
- Purple/blue AI gradients
- Glassmorphism
- Glowing elements
- Excessive shadows

- Excessive rounded cards
 - Generic AI artwork
 - Cryptocurrency-style dashboards
 - Oversized empty hero areas
 - Decorative elements without meaning
 - Excessive animation
-

27. Prototype Data Requirements

Create deterministic demo data for:

- Every Indian state and union territory
- Selected districts
- At least 40 crop scans
- Ten crop options
- Seven prototype conditions
- Healthy and unsupported outcomes
- Weather observations
- Outbreak clusters
- Severity histories
- Verification centers
- Laboratory referrals
- Field findings
- Monthly report
- Retraining events
- Solutions-cache entries

The same case must retain consistent values across all screens.

Random values should not regenerate on every page load.

28. Round 1 Implementation Scope

Component	Round 1 expectation
Public Wellfarm website	Working

Component	Round 1 expectation
Farmer responsive application	Working
Photo upload and validation	Working
GPS permission	Working
Language switching	Working
India-wide officials' dashboard	Working with demo data
State and district drill-down	Working with demo data
Current weather	Live API with fallback
Computer-vision diagnosis	Limited prototype or deterministic fallback
Farmer advisory	Live LLM if configured, with deterministic fallback
Official recommendations	Live or deterministic prototype
Weather–outbreak correlation	Explainable simulated example
Solutions cache	Interactive simulated demonstration
Laboratory directory	Small scraped/curated prototype dataset
Laboratory routing	Real calculation against prototype directory
Laboratory verification	Working prototype workflow
Monthly report	Working preview using demo data
Incremental retraining	Simulated operations panel
Production model deployment	Out of scope
Nationwide field validation	Out of scope

29. Demo Success Criteria

The prototype succeeds if a judge can:

1. Understand Wellfarm's purpose within one minute.
2. Complete the farmer scan journey.
3. See weather and regional context affect the result.
4. Understand uncertainty and responsible-use boundaries.
5. Open the India-wide dashboard.
6. Drill down to a state and district.
7. Understand why a district has its severity.
8. View an official recommendation.

9. Follow a referral to a compatible verification center.
 10. Submit a laboratory/field finding.
 11. See the finding update the collective system.
 12. Understand retraining and caching without being told they are already production systems.
 13. Clearly distinguish live, prototype and simulated data.
 14. Understand why Wellfarm is more than an image-classification application.
-

30. Future Production Scope

Future versions may include:

- Native mobile applications
 - Offline scan submission
 - Broader crop-condition coverage
 - Representative Indian field validation
 - Production laboratory partnerships
 - Government authentication
 - Secure farmer accounts
 - Messaging and notifications
 - Controlled continuous retraining
 - Model registry and rollback
 - Data-drift monitoring
 - Geographic bias monitoring
 - Production reporting
 - Real outbreak feeds
 - Field-survey scheduling
 - Integration with government agricultural systems
-

31. Product Principle

Everything shown as working should genuinely work.

Everything using synthetic or simulated information should be clearly labeled.

The prototype should present a complete, coherent product experience without misrepresenting demo data as real field evidence.